LABORATORY EXPERIMENT

Performance Analyzer Based on Amdahl's Law

Computer Organization & Architecture Laboratory

1. Aim

To develop a performance analyzer based on Amdahl's Law and evaluate the speedup and efficiency of a parallel system for different proportions of parallelizable workloads and different numbers of processors.

2. Problem Statement

Develop a program that implements Amdahl's Law to analyze the performance of a parallel computing system. The analyzer should accept the parallelizable fraction of a workload and different processor counts, calculate theoretical speedup and processor efficiency, determine the maximum theoretical speedup, display the results in tabular form, and compare different workload characteristics.

3. Learning Objectives

- Understand the serial and parallel portions of a program.
- Explain and apply Amdahl's Law.
- Calculate theoretical speedup for different processor counts.
- Calculate processor efficiency.
- Analyze the effect of increasing the number of processors.
- Understand diminishing returns in parallel processing.
- Study the effect of the serial portion on maximum achievable speedup.
- Develop a simple performance-analysis tool using Python.

4. Theory

4.1 Amdahl's Law

Amdahl's Law determines the theoretical speedup obtained when a fraction of a program is parallelized. Let P be the fraction of the workload that can be parallelized, (1 − P) be the serial fraction, and N be the number of processors.

$$S(N) = 1 / [(1 - P) + P/N]$$

4.2 Parallel Efficiency

Efficiency indicates how effectively the available processors are utilized.

$$E(N) = S(N) / N$$

$$\text{Efficiency (\%)} = [S(N) / N] \times 100$$

4.3 Maximum Theoretical Speedup

As the number of processors approaches infinity:

$$S_{max} = 1 / (1 - P)$$

Thus, the serial portion of a program places a fundamental upper bound on speedup.

5. Example Workloads

Workload	Parallelizable Portion (P)	Serial Portion
Workload A	50%	50%
Workload B	75%	25%
Workload C	90%	10%
Workload D	95%	5%
Workload E	99%	1%

6. Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab / Python IDLE
- Matplotlib library

7. Step-by-Step Procedure

1. Open Python IDLE, Jupyter Notebook, or Google Colab and create a new program.
2. Define a function to calculate speedup using $S(N) = 1 / [(1 - P) + P/N]$.
3. Define a function to calculate efficiency using $E = S(N)/N$.
4. Accept or define multiple parallelizable fractions, such as 0.50, 0.75, 0.90, 0.95 and 0.99.
5. Define processor counts such as 1, 2, 4, 8, 16, 32 and 64.
6. For every workload and processor count, calculate theoretical speedup.
7. Calculate efficiency for each processor count.
8. Calculate maximum theoretical speedup using $S_{max} = 1/(1 - P)$.
9. Display workload, parallel fraction, processor count, speedup and efficiency in a table.
10. Generate a Speedup versus Number of Processors graph.
11. Generate an Efficiency versus Number of Processors graph.
12. Compare the workloads and record observations.
13. Verify at least one result manually using the Amdahl's Law formula.

8. Algorithm

1. Start.

2. Read/select the parallelizable fraction P.

3. Calculate the serial fraction as (1 − P).

4. Calculate maximum speedup Smax = 1/(1 − P).

5. Read the processor count N.

6. Calculate S(N) = 1/[(1 − P) + P/N].

7. Calculate E(N) = S(N)/N.

8. Convert efficiency to percentage.

9. Repeat the calculation for all processor counts and workloads.

10. Display the results and generate graphs.

11. Stop.

9. Sample Calculation

Consider a workload in which 90% of the program is parallelizable. Therefore, P = 0.90 and the serial fraction is 0.10. Let N = 8 processors.

S(8) = 1 / [0.10 + (0.90/8)] = 1/0.2125 = 4.7059

Speedup ≈ 4.71×

Efficiency = (4.7059/8) × 100 = 58.82%

10. Sample Laboratory Results: P = 90%

Processors	Speedup	Efficiency (%)
1	1.000	100.00
2	1.818	90.91
4	3.077	76.92
8	4.706	58.82
16	6.400	40.00
32	7.805	24.39
64	8.767	13.70

Maximum theoretical speedup = 1/(1 − 0.90) = 10×.

11. Comparative Results for 16 Processors

Parallel Fraction	Serial Fraction	Speedup	Efficiency
50%	50%	1.882×	11.76%
75%	25%	3.368×	21.05%
90%	10%	6.400×	40.00%
95%	5%	9.143×	57.14%
99%	1%	13.913×	86.96%

12. Maximum Speedup Comparison

Parallel Fraction	Serial Fraction	Maximum Theoretical Speedup
50%	50%	2×
75%	25%	4×
90%	10%	10×
95%	5%	20×
99%	1%	100×

13. Expected Graphical Observations

The Speedup versus Processor Count graph should rise rapidly initially and then approach a limiting value. The Efficiency versus Processor Count graph should generally decrease as the number of processors increases. The curves for workloads with higher parallelizable fractions should show substantially better scalability.

14. Experimental Observations

- With one processor, speedup is 1 and efficiency is 100%.
- Increasing processor count increases speedup, but not linearly.
- Efficiency generally decreases as processor count increases.
- A higher parallelizable fraction produces greater speedup.
- The serial portion limits maximum achievable speedup.
- Adding processors beyond a certain point produces diminishing returns.
- Reducing the serial fraction is critical for scalable parallel performance.

15. Result

The Amdahl's Law Performance Analyzer was successfully developed and used to evaluate theoretical speedup and processor efficiency for different parallelizable workloads and processor counts. The experiment demonstrated that achievable speedup is fundamentally limited by the serial portion of a workload and that increasing the number of processors produces diminishing returns when the parallelizable fraction is limited.